Pakistan Inflation Tracker

An End-to-End Medallion Architecture Pipeline for Weekly Commodity Price Analytics in Pakistan

Data Analysis & Visualization - Semester Project

Phase 1: Project Proposal

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1. Introduction & Objective

Governments and households alike track the cost of living to make short-term decisions - from monetary policy to grocery budgeting. Pakistan's Bureau of Statistics (PBS) publishes a weekly Sensitive Price Indicator (SPI) that captures exactly this signal at a granularity most public inflation dashboards ignore: city-by-city, item-by-item, every week. This project builds an automated, distributed data pipeline that ingests this weekly public release, processes it through the Bronze-Silver-Gold Medallion architecture using Apache Spark on Databricks Community Edition, and surfaces it as a business-intelligence dashboard answering a simple question that changes every week: where, and on what, is inflation actually hitting people right now.

2. Domain & Source Identification

2.1 Project Concept

Domain: Public economic / cost-of-living statistics. The project analyzes retail price movements of 51 essential commodities (food staples, fuel, utilities, and household items) across 17 major urban centres of Pakistan, tracked weekly by a national statistical authority.

2.2 Data Source

Primary source: Pakistan Bureau of Statistics (PBS) - Weekly Sensitive Price Indicator (SPI), Base Year 2015-16=100.

- Landing / index page: https://www.pbs.gov.pk/price-statistics/

Each week, PBS publishes an 'Annexure' file - the city-by-item price matrix across all 17 urban centres - as an Excel download. This is the sole raw file this project ingests, since it is the only PBS release that carries a city dimension at all (PBS also publishes a separate national-level 'Report' summarizing one average price per item for the whole country, but that file has no city breakdown and is out of scope for this pipeline).

- Example, Annexure (city-wise), week ended 24-09-2026: https://www.pbs.gov.pk/wp-content/uploads/2020/07/Annex_24.09.2026.xlsx

Note: file names are not perfectly standardized across weeks (occasional typos, inconsistent hyphenation, and shifting numeric prefixes on some files). The ingestion job therefore parses the actual download links from the PBS Price Statistics page for each week rather than constructing filenames from a fixed date pattern - this is a deliberate design decision, not an oversight, and is documented further in Section 5.

2.3 Ingestion Pattern

- **Full Load:** One-time bulk download of 52 weekly Annexure files covering the last 12 months (approx. Sept 2025 – Sept 2026), establishing the historical baseline.
- **Incremental Load:** A new Annexure file is published every week (typically Thursday/Friday). The pipeline polls the Price Statistics index page weekly, detects the newest "week ended on" date not yet ingested, and appends it - a natural, source-provided incremental cadence with no upserts or deletes to simulate.

3. Data Samples & Volume

3.1 Sample Files Acquired

The following real sample files have been downloaded and will be pushed to the project GitHub repository under /data/samples/:

- Full-load sample: 5 consecutive weekly Annexure files - representative of the historical baseline structure.
- Incremental-load sample: 1 additional weekly Annexure file (the most recent week, 24-09-2026) - representative of a single incoming incremental batch.

3.2 Volume & Frequency Estimation

Metric	Estimate	Notes
Size per weekly Annexure file	~40–150 KB (Excel)	51 items × 17 cities, numeric table
Full load (52 weeks)	~3–8 MB total	Well within Databricks Community Edition storage limits
Incremental load	1 file/week, same size range	Negligible marginal storage cost
Ingestion frequency	Weekly (Thu/Fri)	Matches PBS's own publication cadence

Note: this dataset is intentionally small in raw byte-volume - the value for the course project comes from correctness of the Medallion transformations and dimensional modeling (city × item × week), not from raw scale. This keeps the project comfortably inside Databricks Community Edition's free-tier compute and storage limits (see Section 6).

4. Security & Compliance

4.1 PII Assessment

The source data contains no Personally Identifiable Information. PBS's SPI is a government-published, city-aggregated commodity price index - it contains item names, units, city names, and average prices collected by government price enumerators from open markets. It does not include names, contact details, addresses, IP addresses, financial account details, or any respondent-level information of any individual or business (no shop names, vendor identities, or enumerator identities are disclosed in the public release).

4.2 Handling Strategy

Because no PII exists in this dataset, no masking, hashing, or column-dropping is required for privacy purposes. Standard data-governance hygiene will still be applied: source URLs and retrieval timestamps are logged in Bronze for provenance and auditability, and city names will be standardized against a controlled dimension list (Dim_City) in Silver to prevent inconsistent spellings from silently fragmenting the data.

5. High-Level Medallion Data Modeling

5.1 Bronze Layer (Raw)

Bronze ingests each week's Annexure Excel file as-is, landed into a partitioned path (e.g. /bronze/spi/week_ending=YYYY-MM-DD/), with two metadata columns added: source_url and ingestion_timestamp. No transformation is applied beyond the raw read - this preserves an exact, replayable copy of what PBS actually published.

5.2 Silver Layer (Cleansed)

Silver applies the following cleaning and casting steps:

- Parse the Excel's merged-header, footnote-heavy layout into a tidy row-per-(city, item, week) schema.

- Cast week-ending dates from PBS's DD.MM.YYYY text format to a proper ISO date type.
- Cast price columns from text/mixed-formatted numbers to decimal type; standardize units (e.g. "1 Kg", "Per Litre", "Per Dozen") into a controlled unit dimension.
- Standardize city name spellings against a controlled Dim_City list (PBS's 17 urban centres).
- Deduplicate any week accidentally re-ingested (idempotent upsert keyed on week_ending + city + item).
- Flag and quarantine any week where the source schema deviates unexpectedly (e.g. an item added/removed from the 51-item basket), rather than silently dropping columns.

Planned Silver table: silver.spi_prices_clean - one row per (week_ending_date, city, item_id), with columns: item_name, unit, price_pkr, and the source metadata columns carried from Bronze.

5.3 Gold Layer (Business-Ready)

Gold models the data for direct BI consumption using a star schema:

- Dim_City - 17 urban centres.
- Dim_Item - 51 essential items, tagged with a food / non-food category.
- Dim_Date - ISO week-ending calendar dimension.
- Fact_SPI_Price - the grain table: one row per (date, city, item), with price, week-on-week % change, and year-on-year % change.
- Gold_City_Weekly_Index - aggregated composite price index per city per week, for the city-comparison dashboard.
- Gold_Item_Volatility - rolling standard deviation / max swing per item over the trailing 12 weeks, for the "most volatile commodities" view.

6. Business Intelligence & Dashboards

The final dashboard (Power BI) is designed to answer three concrete business questions:

- Which city is experiencing the sharpest cost-of-living increase this month?
    - a city-ranked heatmap of the Gold_City_Weekly_Index, filterable by date range.
- Which commodities are driving that change?
    - a ranked bar chart of Gold_Item_Volatility, split by food vs. non-food, showing which items moved the most this week.
- How does this week compare, historically?
    - a combined SPI trend line (week-on-week and year-on-year %) over the full 12-month load window, with a reference line at 0% to make inflation/deflation weeks immediately visible.

7. Engineering Setup & FinOps

7.1 Version Control

GitHub repository:

- Repository link: https://github.com/abdulahad-ds/mehngai_metrics

Sample raw files (Section 3.1) are pushed under /data/samples/, notebooks/pipeline code under /src/, and this proposal under /docs/.

7.2 Platform

Databricks Community Edition (free tier) is the target platform for Spark-based processing across all three phases. This is chosen over Azure for Students because the dataset's small volume (Section 3.2) does not need Azure's paid-tier scale, and Community Edition avoids any risk of exhausting a limited student credit balance mid-semester.

7.3 Cost / FinOps Awareness

- Given the small file sizes involved, the single-node Community Edition cluster (15 GB RAM) is sufficient for all three Medallion layers without needing a multi-node cluster.
- The pipeline job is scheduled to run only once per week (matching the source's own cadence), rather than continuously, to avoid unnecessary cluster uptime.
- Notebooks are developed and tested against the small sample files first (Section 3.1) before being pointed at the full 52-week backfill, to avoid burning compute time on debugging against the full dataset.
- Community Edition clusters auto-terminate after inactivity, which is left enabled rather than overridden, as an additional safeguard against idle compute cost.